VASIREDDY VENKATADRI
INSTITUTE OF TECHNOLOGY

Design and Performance Analysis of an Optimized Reversible Quantum Adder-Cum-Subtractor Using Qiskit

Major Project Review - 1

Batch ID: CSM-B07

Project Guide

Mr. B. Naga Raju
Professor, CSM, VVIT

Project Batch Members

Student 1 – 23BQ1A42B6
Student 2 – 23BQ1A42C9
Student 3 – 24BQ5A4207
Student 4 – 23BQ1A4279

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Problem statement

•Conventional arithmetic circuits are generally **irreversible**, which may cause information loss during computation.
•Reversible logic performs computation while **preserving input information**.
•Reversible circuits are useful in areas such as **quantum computing and low-power computing**.
•Existing studies have developed reversible adders, subtractors and approximate arithmetic circuits.
•However, there is scope for a **single optimized circuit** that can perform both addition and subtraction.
•Our project focuses on designing this circuit using **Qiskit** and reducing unnecessary circuit resources.
•The performance will be analyzed using parameters such as **Gate Count, Quantum Cost and Circuit Depth**.
•The circuit will also be Apply image addition and use subtraction for image recovery.
•**Main Problem**

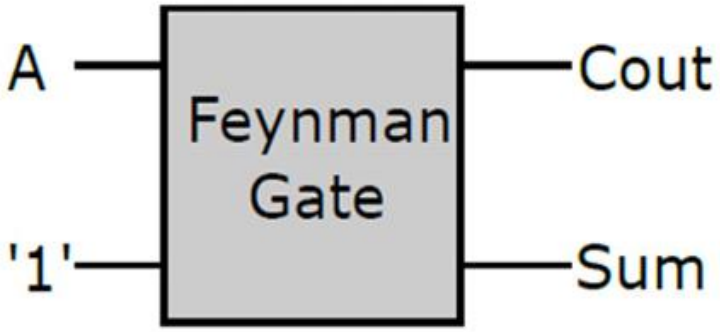
How can we design an optimized reversible quantum adder-cum-subtractor that performs both addition and subtraction while reducing circuit resources?

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

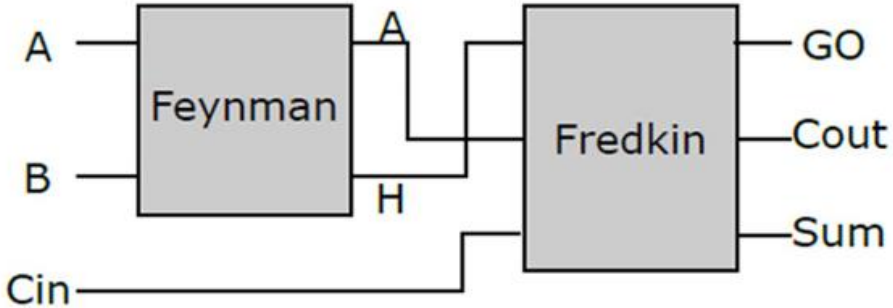
Literature Review **Ref 01 –**Reversible Quantum Circuits for Approximate Computing with Image Processing Application
S. Nitya, A. Khan, M. C. Parameshwara and M. Nagabhushanam, 2025

Proposed Method

- The paper focuses on **reversible approximate computing**.
- It introduces four new **Reversible Approximate Full Adders (RAFAs)**.
- Approximation is introduced in the **Sum and Carry outputs**.
- The main purpose is to reduce the required **circuit resources**.
- Different reversible designs are compared based on their circuit parameters.
- The paper studies **Gate Count (GC), Quantum Cost (QC), Ancilla Inputs (AI) and Garbage Outputs (GO)**.
- The proposed circuits are applied to **image processing**.
- Image quality is checked using **PSNR and SSIM**.



(a) RAFA3



(d) RAFA6

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review **Ref 01 –** Reversible Quantum Circuits for Approximate Computing with Image Processing Application
S. Nitya, A. Khan, M. C. Parameshwara and M. Nagabhushanam, 2025

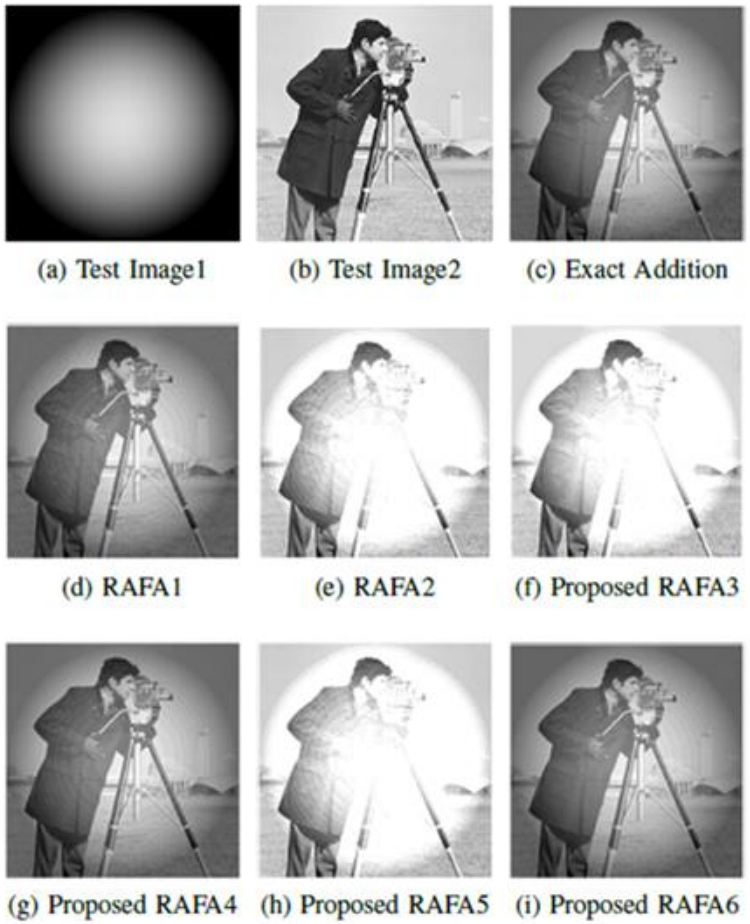
Data & Setup

Experimental Setup

- Reversible approximate full-adder circuits were designed and analyzed.
- **Cadence Xcelium** was used for circuit verification.
- **MATLAB** was used for image-processing experiments.
- Image addition was performed using the reversible circuits.
- Different image outputs were generated for comparison.
- Image quality was measured using:
  - **PSNR – Peak Signal-to-Noise Ratio**
  - **SSIM – Structural Similarity Index**
- Circuit resource parameters were also compared.

Basic Flow

Reversible Circuit → Circuit Verification → Image Addition → Image Quality Analysis



Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review **Ref 01 –**Reversible Quantum Circuits for Approximate Computing with Image Processing Application
S. Nitya, A. Khan, M. C. Parameshwara and M. Nagabhushanam, 2025

Key Results

Circuit Results

- The paper shows that approximation can reduce the required circuit resources.
- **RAFA3** achieved:
  - **50% reduction in Quantum Cost**
  - **66.67% reduction in Gate Count**
- These results show the possibility of designing **smaller reversible circuits**.

Image Processing Results

- **RAFA4** achieved:
  - **PSNR = 34.20 dB**
  - **SSIM = 0.96**
- It reported **7.65% PSNR improvement** and **9.09% SSIM improvement** compared with RAFA1.
- The results demonstrate that resource reduction can be combined with acceptable image quality.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review

Ref 01 – Reversible Quantum Circuits for Approximate Computing with Image Processing Application
S. Nitya, A. Khan, M. C. Parameshwara and M. Nagabhushanam, 2025

Strengths

- Reduces circuit complexity using approximate reversible logic.
- Provides quantitative comparison of different circuit designs.
- Considers important circuit parameters such as **GC and QC**.
- Tests the circuits using actual image-processing applications.
- Uses **PSNR and SSIM** to evaluate image quality.

Limitations

- Mainly focuses on **approximate full-adder circuits**.
- The image-processing application mainly demonstrates **image addition**.
- It does not focus on a **combined adder-cum-subtractor**.
- The main implementation is not based on **Qiskit**.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review **Ref 01 –** Reversible Quantum Circuits for Approximate Computing with Image Processing Application
S. Nitya, A. Khan, M. C. Parameshwara and M. Nagabhushanam, 2025

Relevance to My Research

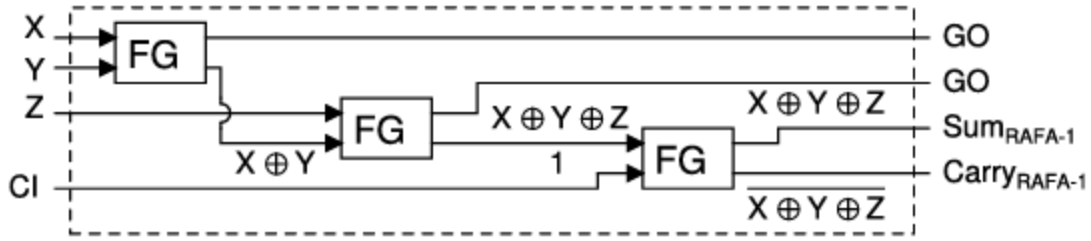
- This paper gives us the idea of **optimizing reversible arithmetic circuits**.
- We can use similar resource parameters for our performance analysis.
- **Gate Count and Quantum Cost** can be used to compare our circuit.
- The use of **PSNR and SSIM** is useful for our image-processing validation.
- Unlike this paper, our work will focus on a **single adder-cum-subtractor**.
- Our circuit will be implemented and tested using **Qiskit**.
- We will extend image processing to **both image addition followed by subtraction-based recovery of the original image.**

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

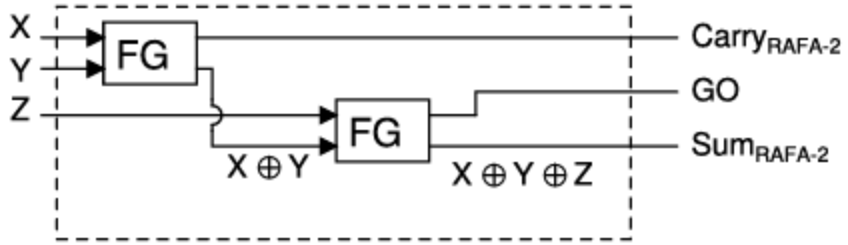
Literature Review **Ref 02** –An Approximate Low-Power Lifting Scheme Using Reversible Logic
S. Raveendran, P. J. Edavoor, N. Y. B. Kumar and M. H. Vasantha, 2020

Proposed Method

- The paper uses **reversible logic** for low-power image processing.
- It proposes two approximate reversible full adders called **RAFA-1 and RAFA-2**.
- Approximation is used to reduce circuit complexity and power requirements.
- The proposed adders are incorporated into a **Haar wavelet lifting scheme**.
- The design is evaluated using reversible-circuit parameters.
- The paper considers **Quantum Cost, Gate Count, Constant Inputs and Garbage Outputs**.
- Image quality is also considered to check the effect of approximation.



(a) Architecture for proposed RAFA-1



(a) Architecture for proposed RAFA-2

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review **Ref 02 –** An Approximate Low-Power Lifting Scheme Using Reversible Logic
S. Raveendran, P. J. Edavoor, N. Y. B. Kumar and M. H. Vasantha, 2020

Data & Setup

- Reversible approximate full-adder circuits were used in the proposed lifting scheme.
- Two designs, **RAFA-1 and RAFA-2**, were evaluated.
- The design was functionally verified using an **FPGA**.
- A **512 × 512 image** was used for the experiment.
- Image quality was measured using **PSNR and SSIM**.
- Circuit parameters were analyzed to study resource requirements.

Evaluation Parameters

Gate Count | Quantum Cost | Constant Inputs | Garbage Outputs | PSNR | SSIM

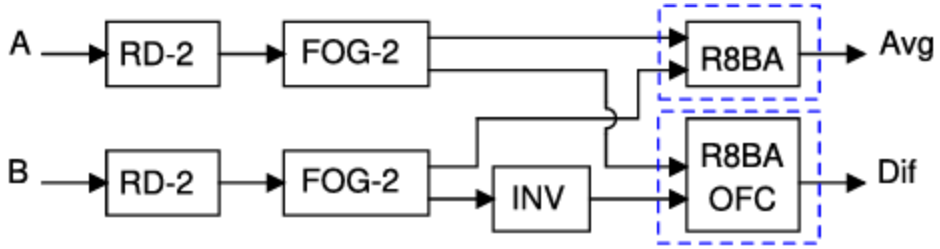


FIGURE 6. Proposed architecture for one level forward Haar wavelet filter using reversible logic.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review **Ref 02 –** An Approximate Low-Power Lifting Scheme Using Reversible Logic
S. Raveendran, P. J. Edavoor, N. Y. B. Kumar and M. H. Vasantha, 2020

Key Results

RAFA-1

- Average **SSIM = 0.9679**
- Average **PSNR = 31.81 dB**

RAFA-2

- Average **SSIM = 0.9696**
- Average **PSNR = 32.15 dB**

Observation

- Both designs produced acceptable image quality.
- The results show that reversible approximate arithmetic can be used while maintaining useful image quality.
- The paper demonstrates the relationship between **resource reduction and image quality**.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review

Ref 02 – An Approximate Low-Power Lifting Scheme Using Reversible Logic
S. Raveendran, P. J. Edavoor, N. Y. B. Kumar and M. H. Vasantha, 2020

Strengths

- Uses reversible logic for low-power computation.
- Focuses on reducing circuit resources.
- Tests the design using an actual image.
- Uses FPGA-based functional verification.
- Uses PSNR and SSIM for image-quality evaluation.

Limitations

- Uses **approximate arithmetic**.
- Main application is a **lifting/wavelet scheme**.
- Does not focus on a combined quantum adder-subtractor.
- The implementation is based on **FPGA**, not Qiskit.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review **Ref 02 –** An Approximate Low-Power Lifting Scheme Using Reversible Logic
S. Raveendran, P. J. Edavoor, N. Y. B. Kumar and M. H. Vasantha, 2020

Relevance to My Research

- The paper supports the use of **reversible logic for resource-efficient computation**.
- Its circuit parameters are useful for our performance analysis.
- Its use of **PSNR and SSIM** is relevant to our image-processing application.
- We will apply similar analysis to our quantum arithmetic circuit.
- Our work will **combined adder-subtractor, Qiskit implementation, image addition, and subtraction-based image recovery.**
- We will implement the circuit using **Qiskit**.
- Image addition and image subtraction will be used for validation.

Literature Review

Ref 03 – Multi-Layer QCA Reversible Full Adder-Subtractor Using Reversible Gates for Reliable Information Transfer and Minimal Power Dissipation on Universal Quantum Computer

Jun-Cheol Jeon, 2024

Proposed Method

- The paper proposes a **reversible full adder-subtractor circuit**.
- The same circuit can perform both **addition and subtraction**.
- Reversible gates are used to preserve information.
- **Feynman gates** are also used in the design.
- The circuit is implemented using a **multi-layer QCA structure**.
- The main aim is to achieve reliable computation with reduced **area, delay and energy dissipation**.

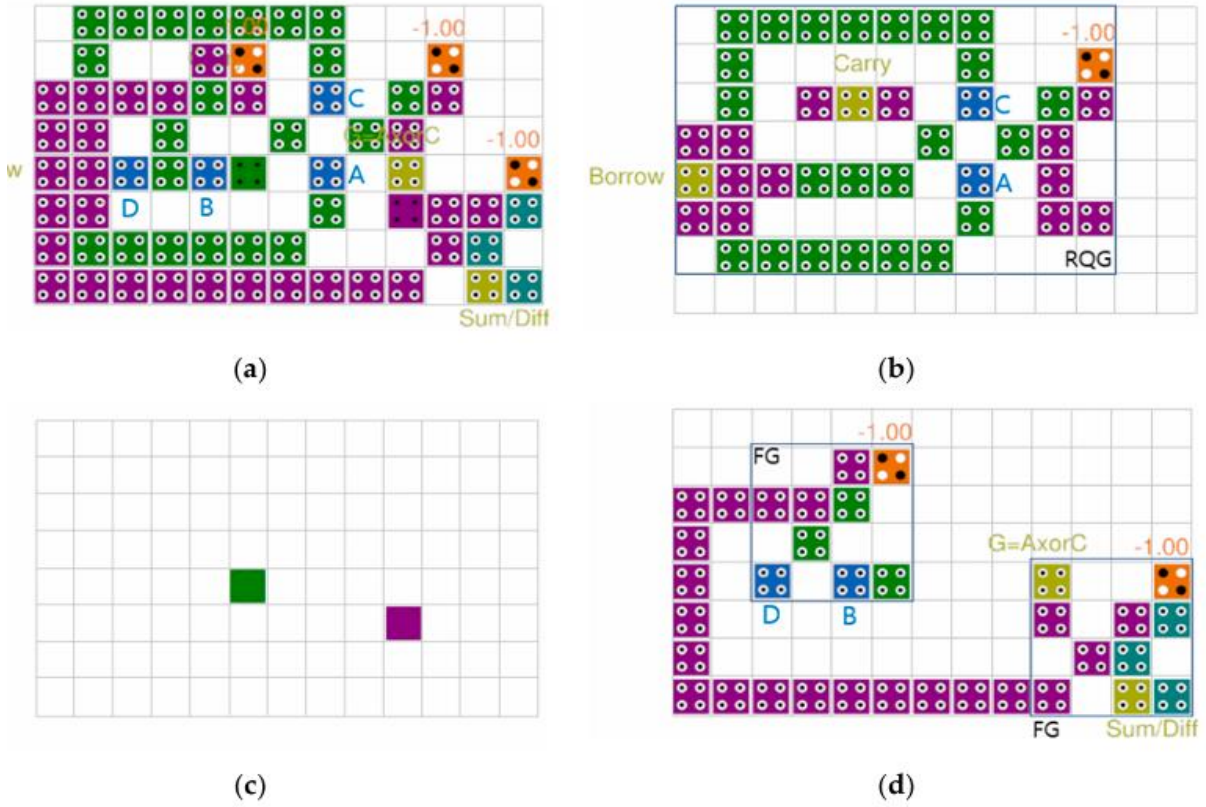


Figure 8. QCA implementation of the proposed RFAS: (**a**) top view; (**b**) first layer; (**c**) second layer; (**d**) third layer.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review

Ref 03 – Multi-Layer QCA Reversible Full Adder-Subtractor Using Reversible Gates for Reliable Information Transfer and Minimal Power Dissipation on Universal Quantum Computer

Jun-Cheol Jeon, 2024

Data & Setup

- The proposed circuit is designed using **Quantum-dot Cellular Automata (QCA)**.
- Reversible gates and Feynman gates are used.
- The design was evaluated using:
- **QCADesigner 2.0.3**
- **QCADesigner-E 2.2**
- The proposed circuit consists of:
- **1 reversible QCA gate**
- **2 Feynman gates**

Performance Parameters

Area | Delay | Energy Dissipation | Output Polarization

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review

Ref 03 – Multi-Layer QCA Reversible Full Adder-Subtractor Using Reversible Gates for Reliable Information Transfer and Minimal Power Dissipation on Universal Quantum Computer

Jun-Cheol Jeon, 2024

Key Results

- The proposed circuit successfully performs **both addition and subtraction**.
- The design uses a small number of reversible components.
- The proposed structure contains **one reversible QCA gate and two Feynman gates**.
- The circuit was analyzed using **area, delay and energy**.
- The work demonstrates that a combined reversible **adder-subtractor** can be designed in a compact structure.
- This combined-operation concept is directly useful for our project.

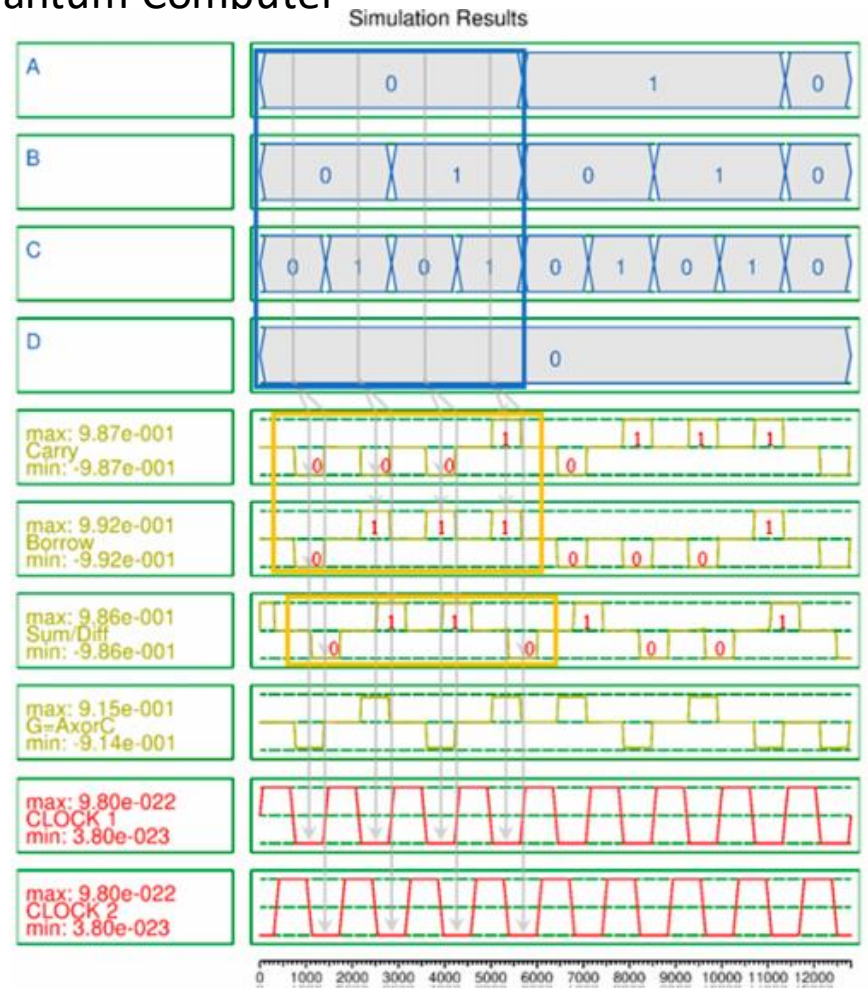


Figure 9. Simulation results of the proposed RFAS circuit.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review

Ref 03 – Multi-Layer QCA Reversible Full Adder-Subtractor Using Reversible Gates for Reliable Information Transfer and Minimal Power Dissipation on Universal Quantum Computer
Jun-Cheol Jeon, 2024

Strengths

- Performs addition and subtraction using one circuit.
- Uses reversible logic.
- Considers area and delay.
- Also analyzes energy dissipation.
- Provides a direct design approach for an adder-subtractor.

Limitations

- The design is based on **QCA technology**.
- Uses **QCADesigner** tools.
- It is not implemented using Qiskit.
- Image processing is not the main application.
- Further optimization for quantum-circuit implementation is possible.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review

Ref 03 – Multi-Layer QCA Reversible Full Adder-Subtractor Using Reversible Gates for Reliable Information Transfer and Minimal Power Dissipation on Universal Quantum Computer
Jun-Cheol Jeon, 2024

Relevance to My Research

- This paper provides the main idea of a **combined reversible adder-subtractor**.
- We will use this concept as a basis for our adder-cum-subtractor.
- Our project will focus on **quantum-circuit implementation using Qiskit**.
- We will try to reduce unnecessary gates and circuit resources.
- We will analyze **Gate Count, Quantum Cost, Circuit Depth and gate operations**.
- The circuit will be tested for **image addition and subtraction-based image recovery.**
- This helps connect the combined arithmetic circuit with our image-processing application.

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Literature Review - Summary

Reference	Year	Main Idea	Strengths	Limitations
Nitya et al.	2025	Approximate reversible adders for image processing	Resource optimization and image evaluation	Mainly image addition
Raveendran et al.	2020	Approximate reversible logic for low-power image processing	Low power and PSNR/SSIM analysis	No combined adder-subtractor
Jeon	2024	Reversible full adder-subtractor using QCA	Addition and subtraction in one circuit	QCA-based
Our Work	2026	Optimized reversible quantum adder-cum-subtractor using Qiskit	combined addition/subtraction + circuit optimization + image recovery and quality analysis	Implementation in progress

Guide: Mr. B . Naga Raju
Batch ID:B07
RegdNo:23BQ1A42B6, RegdNo:23BQ1A42C9
RegdNo: 24BQ5A4207, Regd No:23BQ1A4279

Review – 1

Questions?